



Gen AI and Agentic AI in Production

Week 2
Day 4

4 Week Journey to Prod-Grade

Week 4: Agentic AI in Production

Multi-Agent
Architecture

Capstone:
Agents on
AWS

Capstone:
Frontend

Observability
Monitoring
Security

Agentic AI
Platforms

Week 3: Azure, GCP, Data Engineering, Vectors, MCP

Cyber AI on
Azure with
MCP

Cyber AI on
GCP with
MCP

Amazon
SageMaker

Data Ingest
S3 Vectors

Data Pipes
with MCP

Week 2: AI Platform Engineering on AWS

AWS
Architecture

The Digital
Twin Mk 2

Amazon
Bedrock

Terraform

Github
Actions

Week 1: SaaS Generative AI in Production

First Gen AI
Deployment

Frontend &
Backend

Auth &
Subscription

Healthcare
SaaS App

Deploy to
AWS

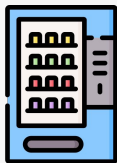
Introducing Terraform

Create AWS services and other infra through code

Controlled



Automated



Repeatable



Checked into Git

.tf files with your configuration
(Not checked in: state files that
map configuration to resources)

Configures everything

No more AWS console!
Everything we did is
automated (except IAM)

Destroy & Recreate

terraform init - stands up your
entire environment
terraform destroy - deletes

AWS CDK is Amazon's proprietary version

We will use Terraform because it works with other providers, too

Terminology

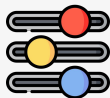
Provider



A vendor like AWS

A plugin to make Terraform apply to that provider

Variable



Parameters

Configurable settings that affect your deployment

Resource



The building block

Each AWS service to be created is described with a resource

State



Record of resources

Maps the configuration to actual resources

Output



Results of deployment

Like, actual CloudFront distribution URLs

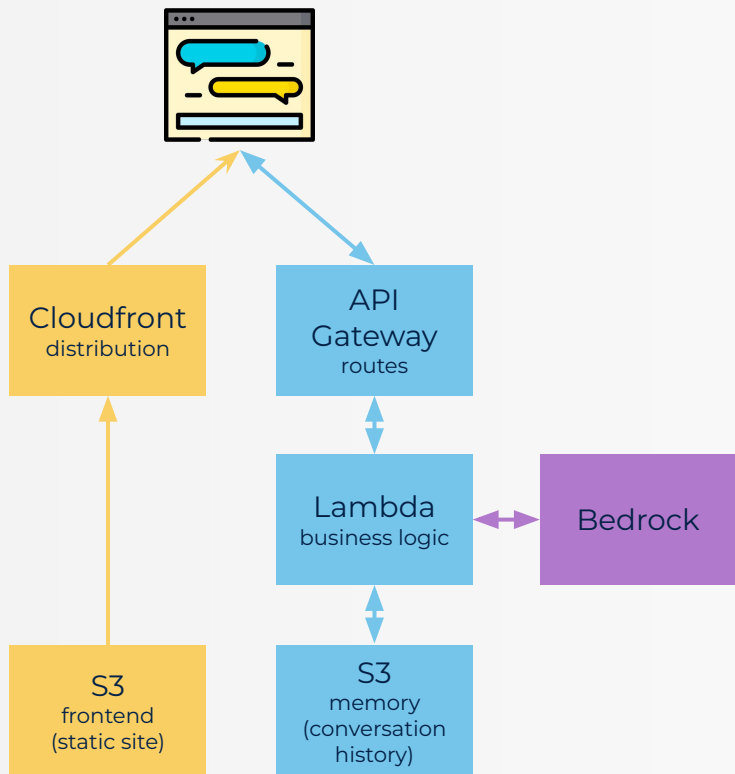
Workspace



Separated state

Isolated state for the same configuration

Deployment Architecture



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45%
Complete